Experiment

05:Problem Statement:-

Write a Python program that the user to enter a year. The program should determine if the year is a leap year or not and print the appropfiate message.

Algorithm:-

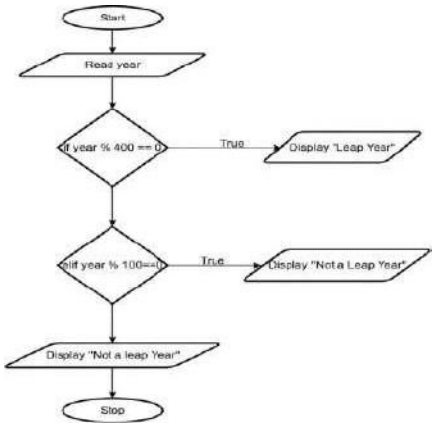
1.Stmt.
2. 2. Read the year from the user
3. 3. If the year is divisible by 400
4. —+ Print "Leap Year"
5. 4. Else if the year is divisible by 100
6. —+ Print "Not a Leap Year" 7. 5. Else if the year is divisible by 4

g. —¥ Print "Leap Year"

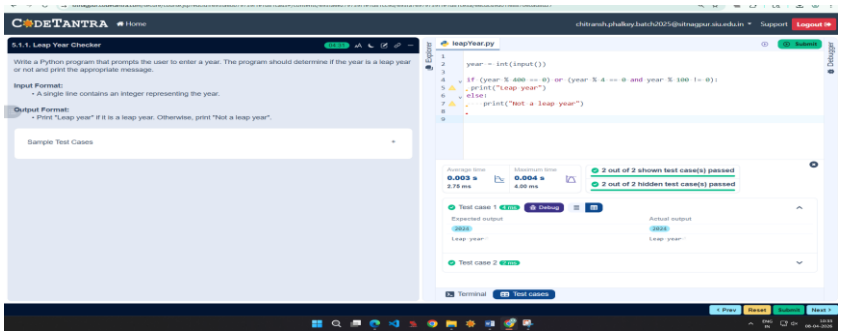
9. 6. Else
10. -+ Print "Not a Leap Year"

11.7. Stop

Flowchart:-



Execution:-



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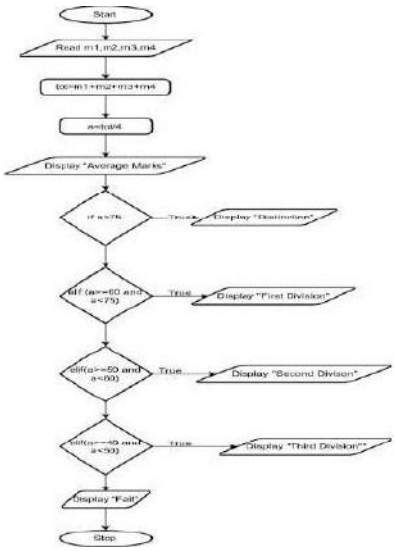
Problem Statement:-

Write a program to calculate the total marks, aggregate percentage, and grade of a student based on marks in four subjects.

Algorithm:-

l. Start
2. Read marks ml, m2, m3, m4
3. total = ml + m2 + m3 + m4
4. percentage = total / 4
5. If percentage > 75 —y Print "Distinction"
6. Else if percentage > 60 —4 Print "First Division"
7. Else if percentage > 50 —Y Print "Second Division'
8. Else if percentage > 40 Print "Third Division"
9. Else Print "Fail"
10. Print total and percentage
11. Stop.

Flowchart:-



Execution:-

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5.1.2. Student Grade Based on Aggregate

Write a program to calculate the total marks, aggregate percentage, and grade of a student based on marks in four subjects. The grade is determined as follows:

- Aggregate >= 75%: Distinction
- Aggregate >= 60% and < 75%: First Division
- Aggregate >= 50% and < 60%: Second Division
- Aggregate >= 40% and < 50%: Third Division
- Aggregate < 40%: Fail

Input Format:

- Four space-separated integers representing the marks in four subjects.

Output Format:

- The first line should print the total marks.
- The second line should print the aggregate percentage with two decimal places.
- The third line should print the grade.

Constraints:

- 0<=marks in each subject<=100

Sample Test Cases

studentG...

```python
marks = list(map(int, input().split()))


total = sum(marks)


aggregate = total / 4


if aggregate >= 75:
    grade = "Distinction"
elif aggregate >= 60:
    grade = "First Division"
elif aggregate >= 50:
```

Average time 0.003 s 2.50 ms

Maximum time 0.004 s 4.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 4 ms Debug

Expected output	Actual output
85 90 78 88	85 90 78 88
341	341
85.25	85.25
Distinction	Distinction

Test case 2 3 ms

Terminal Test cases

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